

Phase Vocoding Demo

Fourier Transform Settings

```
% Frame & Hop Size
N = 128;
HOP_SIZE = N/8;

H = HOP_SIZE / N; % HOP_FACTOR

% Windowing
% WINDOW = ones(N, 1);
WINDOW = kaiser(N, 4);

% Fourier transform
FFT = dsp.FFT();
IFFT = dsp.IFFT();
```

Pitch-Shift Settings

```
% Synthesized Frame & Hop Size
M = N*2;
HOP_SIZE_M = M * H;
WINDOW_M = kaiser(M, 4);

PITCH_SHIFT_FACTOR = M / N;

% Phase locking (CAN and WILL be changed)
FIND_PHASE_LOCK_BINS = @valleyPhaseLock;
```

Graph Settings

```
% Colors
PEAK_BIN_COLOR = [0.522 0.086 0.820];
MUTUAL_PEAK_BIN_COLOR = [1 0 0];
ESTIMATED_FREQUENCIES_COLOR = [0.231 0.667 0.196];
```

Signal Definition

```
% Signal Length (Best if in increments of HOP_SIZE)
SIGNAL_LENGTH = HOP_SIZE*30;

% All frequency terms ( $\omega$ ,  $\Phi$ , A)
FREQUENCIES = [
    9.8/N, 0, 1;
];
```

```
% The signal is the sum of multiple complex sinusoidal terms
x = @(t) ... % Signal
    sum(exp(1j * (2 * pi * FREQUENCIES(:, 1) * t + FREQUENCIES(:, 2))) .*
FREQUENCIES(:, 3), 1);

TOTAL_FRAMES = floor(( SIGNAL_LENGTH - N ) / HOP_SIZE);
```

Pitch-Shifting Code

Loop Over The Entire Signal

```
% Initialization
prevPeaks = [];
prevPhaseSpec = zeros(N, 1);
prevPhaseSpec_Modulated = zeros(M, 1);

outputtedSignal = zeros(SIGNAL_LENGTH / N * M, 1);

% GRAPHING PURPOSES
% Initialize plots for the figures
frameNum = 1;

sampledSignalPlots = gobjects(TOTAL_FRAMES, 1);
windowedSignalPlots = gobjects(TOTAL_FRAMES, 1);
ampSpecPlot = gobjects(TOTAL_FRAMES, 4);
interpolatedAmpSpecPlot = gobjects(TOTAL_FRAMES, 1);
synthesizedFramePlot = gobjects(TOTAL_FRAMES, 2);

% Update
for n = 0 : HOP_SIZE : TOTAL_FRAMES * HOP_SIZE

% Body
```

Sampling and Preprocessing

```
% Sample the signal at t = n
x_sampled = x( n:(n + N-1) )'; % Samples frames n to n + N - 1

% Window the sampled signal
x_windowed = x_sampled .* WINDOW;

% GRAPHING PURPOSES
% Graph the sampled and windowed
figure(1);
sampledSignalPlots(frameNum) = plot(0:N-1, real(x_sampled)', Color="b");
hold on;
```

```
figure(2);
windowedSignalPlots(frameNum) = plot(0:N-1, real(x_windowed)', Color="b");
hold on;
```

Short-Time-Fourier-Transform

```
% STFT is implicit
X = FFT( conj( x_windowed ) ); % Turn into the conjugate so that it transforms
properly

% Convert to polar spectrum
X_polar = convertToPolar( X );

% GRAPHING PURPOSES
% Graph the amplitude spectrum
figure(3);
ampSpecPlot(frameNum, 1) = stem(0:N-1, real(X_polar(:, 1))', Color="blue",
Marker="none");
hold on;
```

Estimate Frequencies

```
% Find the current and mutual peaks
currentPeaks = findPeaks( X_polar(:, 1) );
mutualPeaks = intersect(prevPeaks, currentPeaks);

peakFrequencies = currentPeaks - 1; % Make a list of the frequencies that peak

% Get the phases of the corresponding mutual peaks and calculate phase difference
phaseDifferences = X_polar(mutualPeaks, 2) - prevPhaseSpec(mutualPeaks);

% Calculate frequency deviation and add to the frequency bins
% (mutualPeaks-1) gets the frequency of the mutual peak bins
frequencyDeviations = wrap( phaseDifferences/2/pi/H - (mutualPeaks-1) , 1/H );
estimatedFrequencies = (mutualPeaks-1) + frequencyDeviations;

% Set the mutual peaks to their new estimated frequencies
peakFrequencies(ismember(peakFrequencies, mutualPeaks-1)) = estimatedFrequencies;

% GRAPHING PURPOSES
% Graph peaks and estimated frequency
figure(3);
ampSpecPlot(frameNum, 2:4) = [
    stem(currentPeaks-1, X_polar(currentPeaks, 1), Color=PEAK_BIN_COLOR,
Marker="x"), ...
```

```

    stem(mutualPeaks-1, X_polar(mutualPeaks, 1), Color=MUTUAL_PEAK_BIN_COLOR,
Marker="x"), ...
    stem(estimatedFrequencies, estimatedFrequencies * 0,
Color=ESTIMATED_FREQUENCIES_COLOR, Marker="|", Linewidth=2) ...
];

```

Modulate Spectrum

```

% Interpolate Spectrum (Linear interpolation of amplitude and phase spectrum)
X_polar_interpolated = interpolateSpectrum(X_polar, N, M);

% Scale the peak frequencies to the interpolated spectrum
peakFrequencies = peakFrequencies / N * M;

% Create the spectrum of modulated values
modulatedSpec = X_polar_interpolated;

if n ~= 0
% Iterate through all the frequency spectrums
    for peakFrequency = peakFrequencies'

        % Find the corresponding peaking bin
        peakBin = round(peakFrequency) + 1;

        % Calculate the phase correction
        phaseCorrection = ( ...
            prevPhaseSpec_Modulated(peakBin) ...
            + 2 * pi * H * peakFrequency ...
        ) - modulatedSpec(peakBin, 2);

        % Adjust the phase
        phaseLockRange = FIND_PHASE_LOCK_BINS(peakBin, modulatedSpec(:, 1));
        modulatedSpec(phaseLockRange(1):phaseLockRange(2), 2) =
modulatedSpec(phaseLockRange(1):phaseLockRange(2), 2) + phaseCorrection;
    end
end

% GRAPHING PURPOSES
figure(4);
interpolatedAmpSpecPlot(frameNum) = stem(0:M-1, X_polar_interpolated(:, 1)',
Color="g", Marker="none");
hold on;

```

Resynthesize the Spectrum

```

% Resynthesize
Y = IFFT( convertToComplex( modulatedSpec ) );
Y = conj( Y );

```

```

% Y = Y ./ [WINDOW; WINDOW];

% Properly scale the frame
Y = Y / 2; % I'm too lazy to calculate the actual scale factor

% Overlap add the result
n_M = n / HOP_SIZE * HOP_SIZE_M;
outputtedSignal((n_M + 1):(n_M + M)) = outputtedSignal((n_M + 1):(n_M + M)) + Y;

% GRAPHING PURPOSES
figure(5);
synthesizedFramePlot(frameNum, 1) = plot(real(Y)', "blue");
hold on;
synthesizedFramePlot(frameNum, 2) = plot(real(x_windowed)', "red");

```

Update and End The Loop

```

% Update the previous spectrum information
prevPeaks = currentPeaks;
prevPhaseSpec = X_polar(:, 2);
prevPhaseSpec_Modulated = modulatedSpec(:, 2);

% Increment frame number
frameNum = frameNum + 1;

end

figure(6);
plot(real(outputtedSignal)', "blue");
hold on;
plot(real(x(0:SIGNAL_LENGTH))', "red");
hold off;

ylim([-10, 10]);

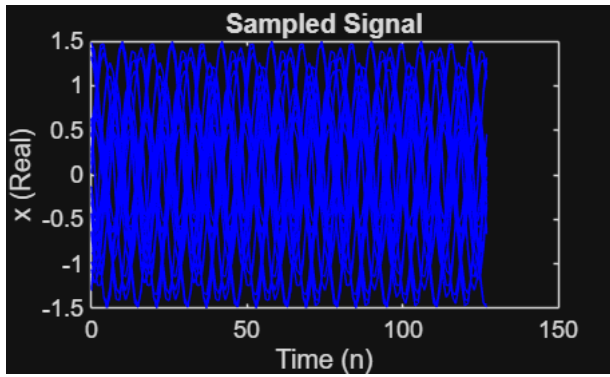
```

Label the figures

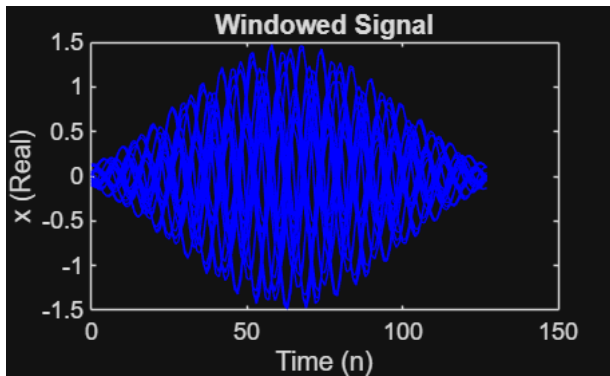
```

%% labelFigure does hold off automatically
labelFigure(1, "Sampled Signal", "Time (n)", "x (Real)");

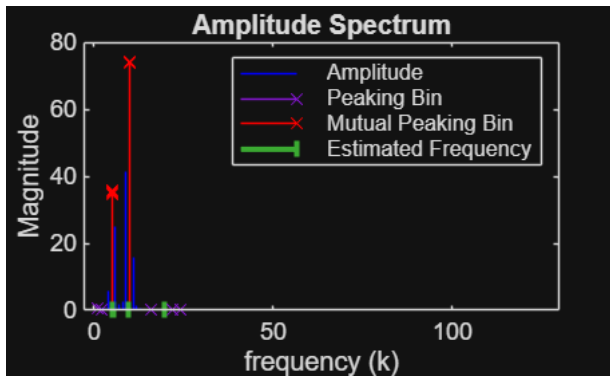
```



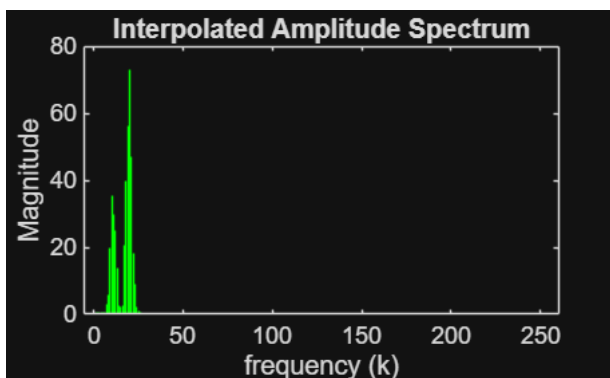
```
labelFigure(2, "Windowed Signal", "Time (n)", "x (Real)");
```



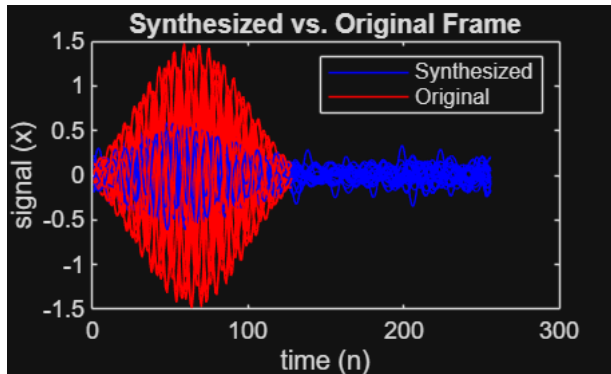
```
labelFigure(3, "Amplitude Spectrum", "frequency (k)", "Magnitude");
legend({"Amplitude", "Peaking Bin", "Mutual Peaking Bin", "Estimated Frequency"});
```



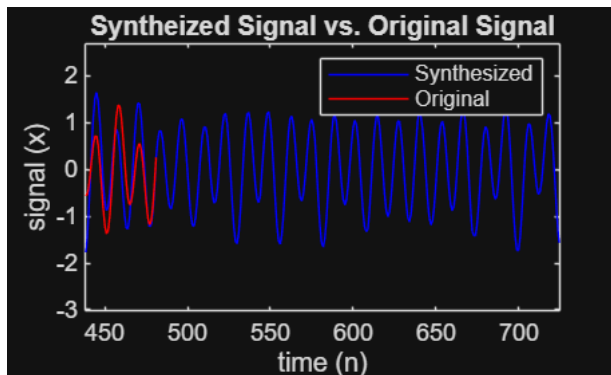
```
labelFigure(4, "Interpolated Amplitude Spectrum", "frequency (k)", "Magnitude");
```



```
labelFigure(5, "Synthesized vs. Original Frame", "time (n)", "signal (x)");
legend({"Synthesized", "Original"});
```



```
labelFigure(6, "Syntheized Signal vs. Original Signal", "time (n)", "signal (x)");
legend({"Synthesized", "Original"});
```



Graph Controls

```
selectedFrame = 5
```

```
selectedFrame =  
5
```

```
selectFrameData(selectedFrame, [ ...  
    sampledSignalPlots, ...  
    windowedSignalPlots, ...  
    ampSpecPlot, ...  
    interpolatedAmpSpecPlot, ...  
    synthesizedFramePlot, ...  
]);
```

